Oligopoly

EC 111 Principles of Microeconomics

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Reading

- These slides are not a substitute for reading Mankiw, Chapter 18, Oligopoly.
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 18, "Oligopoly".

What you must know from this chapter

1. **Conditions:** what makes a market an oligopoly.
2. **Cartels:** firms acting together as one monopoly.
3. **Collusion:** agreements to restrict output or fix prices, and why they break down.
4. How the outcome under oligopoly compares with monopoly and perfect competition.

The model used in this lecture

Two firms, A and B, sell an identical product.

- Market demand is a straight line:

$$P = a - bQ, \qquad Q = q_A + q_B.$$

- q_A and q_B are the two firms' outputs. Q is total output.
- Both firms have the same constant marginal cost, c, and no fixed cost.
- So each unit costs c, and average cost is also c.

Part 1

What Makes a Market an Oligopoly?

Oligopoly

Definition: Oligopoly
A market with only a few sellers, each offering a product similar or identical to the others'.

- **Few sellers:** a handful of firms supply most of the market.
- **Similar or identical products.**
- **Barriers to entry:** new firms find it hard to enter.
- **Interdependence:** each firm's profit depends on what its rivals do.

Examples: commercial aircraft (Boeing and Airbus); U.S. wireless phone service.

Definition: Duopoly
An oligopoly with exactly two sellers.

Interdependence, written as an equation

Firm A's profit is its profit per unit times its output.

$$
\begin{aligned}
\pi_A &= (P - c) \times q_A && \text{profit per unit times units} \\
&= (a - bQ - c) \times q_A && \text{replace } P \text{ with } a - bQ \\
&= (a - b\,q_A - b\,q_B - c) \times q_A && \text{replace } Q \text{ with } q_A + q_B
\end{aligned}
$$

- Firm B's output, q_B, appears in firm A's profit.
- The more B sells, the lower the price, and the less A earns on each unit.

Finding

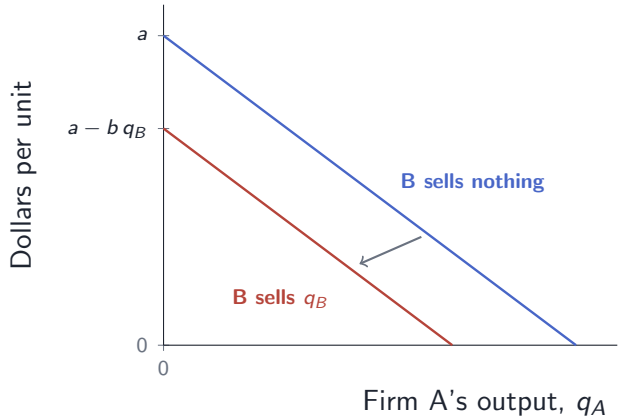
In an oligopoly, a firm's profit depends on its rival's choice. In perfect competition and in monopoly, no rival's choice enters a firm's profit.

The demand left for firm A

- Rearrange demand:

$$P = (a - b\,q_B) - b\,q_A.$$

- For a given q_B, this is the demand left for firm A.
- It starts at $a - b\,q_B$ and has slope $-b$.
- When B sells more, A's demand shifts down.



Four market structures

	Perfect competition	Monopolistic competition	Oligopoly	Monopoly
Sellers	Many	Many	**A few**	One
Product	Identical	Differentiated	**Similar or identical**	No close substitutes
Entry	Free	Free	**Difficult**	Blocked
Rivals matter?	No	Little	**Yes**	No rivals

Finding

What sets oligopoly apart is interdependence: each firm must allow for what its rivals do.

Part 1 summary

- Oligopoly: a few sellers of similar or identical products.
- Entry is difficult.
- Interdependence: firm A's profit depends on firm B's output.
- When B sells more, the demand left for A shifts down.
- Compared with perfect competition and monopoly: there, no rival's choice enters a firm's profit.

Check: Part 1

Which of the following examples illustrates an oligopoly market?

A A farmers' market with many individuals selling sweet corn and tomatoes

B A city whose electrical service is provided by one electric co-operative

C A city with two firms that are licensed to sell school uniforms for the local schools

D A city with many independently owned hair styling salons

Check: Part 1 – answer

Which of the following examples illustrates an oligopoly market?

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Finding

Two sellers of the same product, with entry blocked by licensing: a duopoly. A is perfect competition, B is monopoly, D is monopolistic competition.

Part 2

Cartels: Acting as One Monopoly

Collusion and cartels

Definition: Collusion

An agreement among firms in a market about the quantities to produce or the prices to charge.

Definition: Cartel

A group of firms acting together. A successful cartel acts as one monopolist.

- The cartel chooses total output Q to maximize joint profit.
- Then it splits Q between the members.

The cartel's output: the monopoly rule

As one monopolist, the cartel produces where MR = MC.

$\text{MR} = a - 2bQ$	straight-line demand: MR twice as steep (Chapter 16)
$a - 2bQ = c$	set MR = MC
$2bQ = a - c$	add $2bQ$ to both sides; subtract c
$Q_{\text{cartel}} = \dfrac{a-c}{2b}$	divide both sides by $2b$

Price, from demand:

$$P_{\text{cartel}} = a - b \times \frac{a-c}{2b}$$ put Q_{cartel} into $P = a - bQ$

$$= a - \frac{a-c}{2} = \frac{a+c}{2}$$ the b's cancel

The benchmark: perfect competition

Under perfect competition, price equals marginal cost.

$$
\begin{aligned}
a - bQ &= c && \text{set } P = \text{MC} \\
Q_{\text{comp}} &= \frac{a-c}{b} && \text{solve for } Q \\
P_{\text{comp}} &= c
\end{aligned}
$$

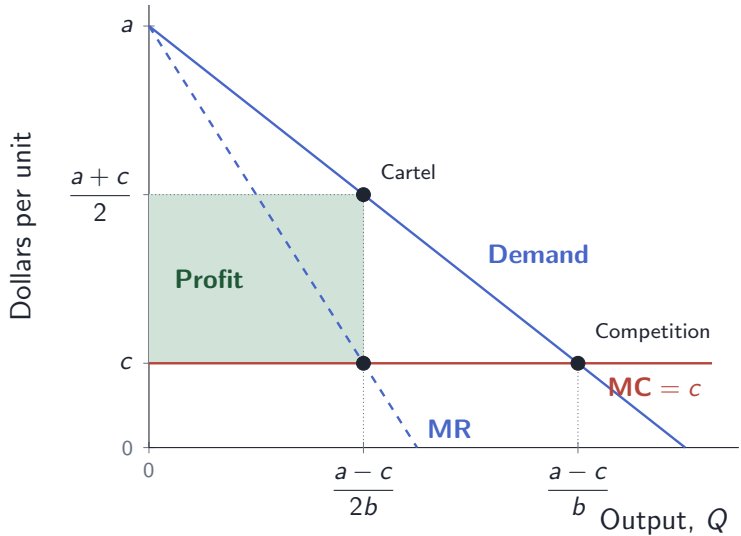
Compare the two outputs:

$$Q_{\text{cartel}} = \frac{a-c}{2b} = \frac{1}{2} \times \frac{a-c}{b} = \frac{1}{2} Q_{\text{comp}}.$$

Finding

A cartel produces half the competitive output and charges $\frac{a+c}{2}$, above marginal cost c.

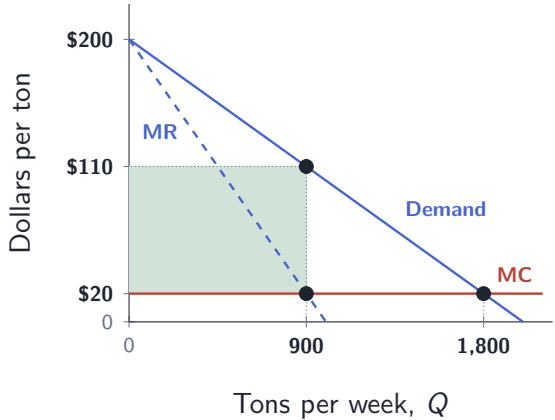
The cartel on the graph



Example: a cartel of two cement firms

$P = 200 - Q/10$, so $a = 200$, $b = 1/10$; $c = \$20$.

1. $Q_{\text{cartel}} = \dfrac{200 - 20}{2 \times 1/10} = 900$ tons.
2. $P_{\text{cartel}} = \dfrac{200 + 20}{2} = \$110$.
3. Profit $= (110 - 20) \times 900 = \$81{,}000$.
4. Split equally: 450 tons and \$40,500 each.
5. Competition: $Q = \dfrac{180}{1/10} = 1{,}800$ tons at \$20.



Part 2 summary

- Collusion: firms agree on quantities or prices.
- A cartel acts as one monopolist: MR = MC.
- $Q_{\text{cartel}} = \dfrac{a-c}{2b}$ and $P_{\text{cartel}} = \dfrac{a+c}{2}$.
- The members split the output and the profit.
- Compared with perfect competition: half the output, and a price above MC.

Check: Part 2

If duopoly firms that are not colluding were able to successfully collude, then

A price and quantity would rise.

B price and quantity would fall.

C price would rise and quantity would fall.

D price would fall and quantity would rise.

Check: Part 2 – answer

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A price and quantity would rise.

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Finding

A cartel restricts output to the monopoly level, so quantity falls and price rises.

Part 3

Why Collusion Breaks Down

The question

- The cartel output gives the two firms the most profit they can earn together.
- Yet cartels break down. Why would a member break an agreement that serves the group best?
- To answer, look at what one more unit of output does to profit.
- First for the cartel as a whole, then for firm A alone.

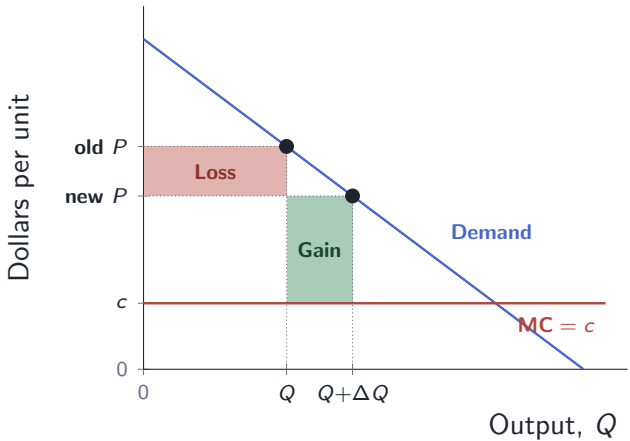
Marginal cost in this model

- Each unit costs the same amount, c, and there is no fixed cost.
- So total cost is $\text{TC} = c \times Q$.
- One more unit adds exactly c to total cost. So marginal cost is $\text{MC} = c$, the same for every unit.
- A unit sold at price P therefore earns $P - c$ over its cost.

One more unit: a gain and a loss

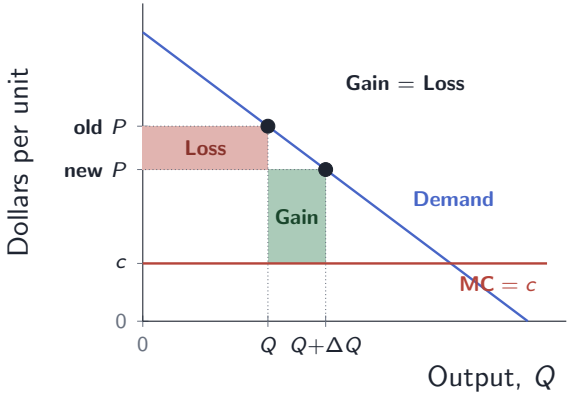
Output rises from Q to $Q + \Delta Q$.
To sell more, the price must fall.

- **Gain (green):** the extra units each earn new $P - c$. This is the output effect.
- **Loss (pink):** the Q units already sold now fetch a lower price. This is the price effect.



At the cartel output, the gain equals the loss

- If the gain were bigger than the loss, more output would raise joint profit. The cartel would produce more.
- If the loss were bigger, less output would raise joint profit. The cartel would produce less.
- So the cartel stops where the two areas are equal.

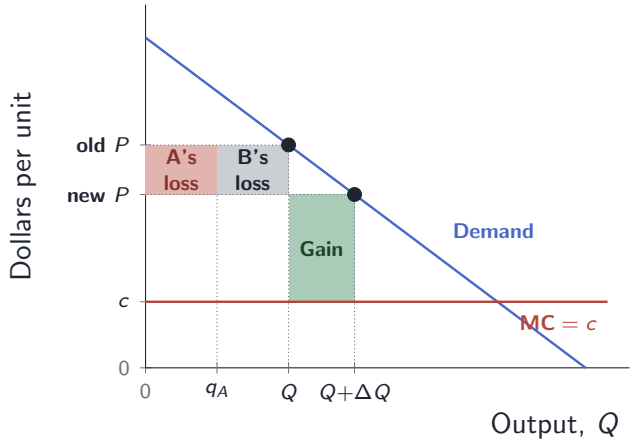


Finding

At the cartel output, one more unit leaves joint profit unchanged. This is the cartel's MR = MC rule, seen as two areas (Appendix A.1).

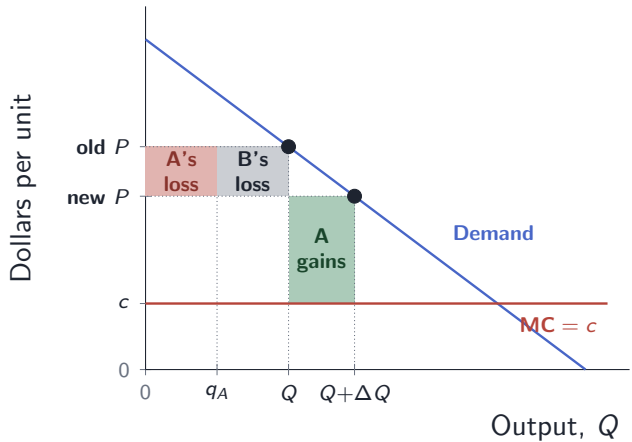
Who bears the loss?

- The cartel's output is split: A sells q_A, B sells q_B.
- The lower price applies to every unit sold, A's and B's alike.
- So the pink loss splits in two: one part on A's units, one part on B's units.



Firm A's view: produce more on its own

- Suppose firm A alone produces the extra units.
- A collects the whole gain (green).
- A bears only the loss on its own units (pink).
- The loss on B's units (grey) falls on B. A ignores it.
- Green equals the whole loss, so green is bigger than pink. A comes out ahead.



Why cartels break down

- For the cartel, the extra output is not worth producing: gain and loss cancel.
- For firm A alone, it is worth producing: A keeps all the gain and pays only part of the loss.
- Firm B sees exactly the same thing from its side.
- So each member wants to produce more than its agreed share.

Finding

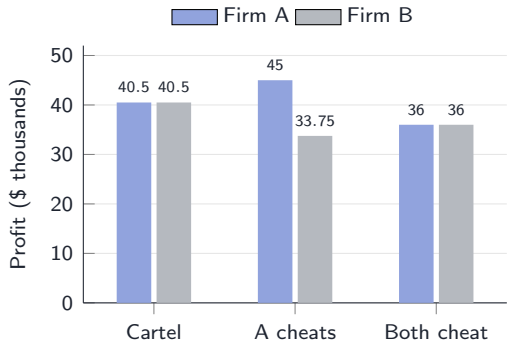
Each member gains by producing more, because part of the cost falls on its partner. That is why cartels break down. In symbols, A's gain is B's share of the loss: $\Delta\pi_A = -\,q_B\,\Delta P > 0$ (Appendix A.1).

Example: the cement cartel

Each firm agreed to 450 tons.

- **Cartel:** \$40,500 each.
- **A cheats** (600 tons; B keeps 450): A earns \$45,000, B \$33,750.
- **Both cheat** (600 each): \$36,000 each.

Working: Appendix A.2.



Finding

Cheating pays the cheat. When both cheat, both end up below the cartel profit.

The payoff table and the dominant strategy

Each firm chooses 450 tons (keep the agreement) or 600 tons (cheat). Profits are (A, B).

	B makes 450	B makes 600
A makes 450	($40,500, $40,500)	($33,750, $45,000)
A makes 600	($45,000, $33,750)	($36,000, $36,000)

- If B keeps to 450, A does better at 600.
- If B makes 600, A still does better at 600.
- So 600 is best for A whatever B does. The same holds for B.

Definition: Dominant strategy

The best choice for a player whatever the other player does. Here: 600 tons, for both firms.

Nash equilibrium and the prisoners' dilemma

Definition: Nash equilibrium

A pair of choices in which each player's choice is its best, given the other's choice. Neither gains by changing its own choice alone.

- At (600, 600), neither firm gains by switching back to 450 on its own.
- So (600, 600) is the Nash equilibrium.
- Yet both would earn more if both kept the agreement.

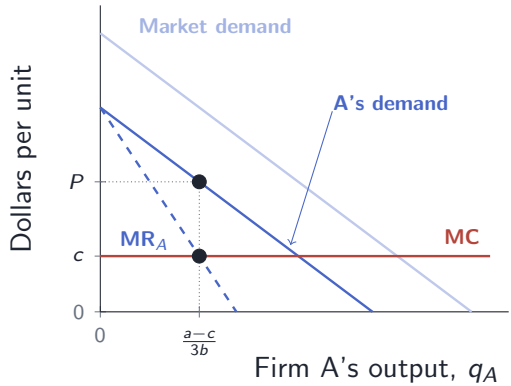
Definition: Prisoners' dilemma

A game in which each player's own best choice leaves both worse off than if they cooperated.

Where the firms end up

- Each firm takes its rival's output as given and sets its own $\text{MR} = \text{MC}$.
- Result: each firm produces $\dfrac{a-c}{3b}$.
- That is more than its cartel share, $\dfrac{a-c}{4b}$.
- Cement: 600 tons each, against 450 in the cartel.

Working: Appendix A.3.



Part 3 summary

- One more unit brings a gain (output effect) and a loss (price effect).
- At the cartel output the gain equals the loss.
- A single member keeps all the gain but bears only its own part of the loss.
- So each member gains by producing more, and the cartel breaks down.
- Dominant strategy: produce more. Nash equilibrium: both produce more.
- Prisoners' dilemma: both end up with less profit than under the cartel.

Check: Part 3

Cartels are difficult to maintain because

A the monopoly output is very difficult to determine.

B the number of firms is always large.

C costs to the firms in a cartel are continually rising.

D each firm has an incentive to deviate from its agreed output level.

Check: Part 3 – answer

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A the monopoly output is very difficult to determine.

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Finding

At the cartel output, $\Delta\pi_A = -q_B \Delta P > 0$: each member gains by producing more.

Part 4

Market Outcomes Compared

Three outcomes from the same demand and cost

Structure	Output	Price	Total profit	P and MC
Monopoly or cartel	$\frac{a-c}{2b}$	$\frac{a+c}{2}$	Highest	$P > \text{MC}$
Oligopoly, two firms	$\frac{2(a-c)}{3b}$	$\frac{a+2c}{3}$	Lower	$P > \text{MC}$
Perfect competition	$\frac{a-c}{b}$	c	Zero	$P = \text{MC}$

The oligopoly price, from demand: $P = a - b \times \frac{2(a-c)}{3b} = a - \frac{2(a-c)}{3} = \frac{a+2c}{3}$.

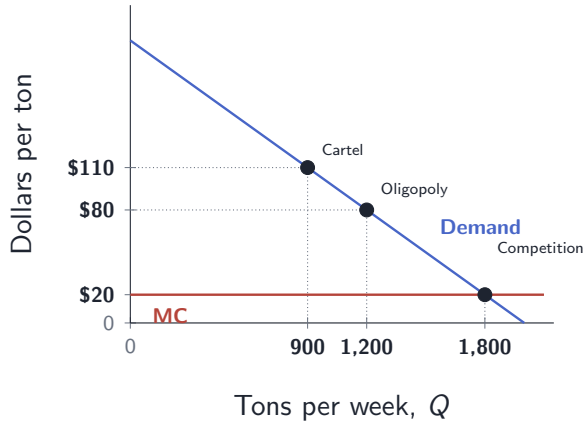
Finding

Oligopoly output lies between monopoly and perfect competition, and so does its price.

The cement market: three outcomes

	Q	P
Cartel	900	\$110
Oligopoly	1,200	\$80
Competition	1,800	\$20

Total profit: \$81,000, then
$2 \times 36{,}000 = \$72{,}000$, then \$0.



All four market structures in the long run

	Perfect competition	Monopolistic competition	Oligopoly	Monopoly
Price and MC	$P = \text{MC}$	$P > \text{MC}$	$P > \text{MC}$	$P > \text{MC}$
Long-run profit	Zero	Zero	Can be positive	Can be positive
Why	Free entry	Free entry	Entry is difficult	Entry is blocked
Output	Efficient	Below efficient scale	Below competitive	Lowest

Finding

Only perfect competition sets $P = \text{MC}$. In the other three, some buyers who value a unit above its cost do not get it.

Collusion and the law

- Collusion raises price and cuts output, so buyers lose.
- **Sherman Antitrust Act (1890):** agreements among competitors to fix prices or restrict output are illegal.
- **Clayton Act (1914):** individuals harmed by illegal agreements can sue to recover damages.
- **Example:** in 1996 Archer Daniels Midland agreed to plead guilty to fixing the prices of lysine and citric acid, two food and feed additives, and paid a $100 million fine.

Source: U.S. Department of Justice, press release, 15 October 1996.

Part 4 summary

- Cartel: output $\frac{a-c}{2b}$, the lowest; price $\frac{a+c}{2}$, the highest.
- Oligopoly with two firms: output $\frac{2(a-c)}{3b}$; price $\frac{a+2c}{3}$.
- Perfect competition: output $\frac{a-c}{b}$; price c.
- Oligopoly lies between monopoly and perfect competition.
- Price-fixing is illegal under the Sherman Act.

Check: Part 4

The equilibrium quantity in markets characterized by oligopoly is

A higher than in monopoly markets and higher than in perfectly competitive markets.

B higher than in monopoly markets and lower than in perfectly competitive markets.

C lower than in monopoly markets and higher than in perfectly competitive markets.

D lower than in monopoly markets and lower than in perfectly competitive markets.

Check: Part 4 – answer

The equilibrium quantity in markets characterized by oligopoly is

A higher than in monopoly markets and higher than in perfectly competitive markets.

B higher than in monopoly markets and lower than in perfectly competitive markets.

C lower than in monopoly markets and higher than in perfectly competitive markets.

D lower than in monopoly markets and lower than in perfectly competitive markets.

Finding

$\frac{a-c}{2b} < \frac{2(a-c)}{3b} < \frac{a-c}{b}$. In cement: $900 < 1{,}200 < 1{,}800$.

Which rule answers which question

Question	Rule
What defines an oligopoly?	A few sellers, similar products, difficult entry, interdependence.
What does a cartel produce?	The monopoly output: MR = MC, $Q = \frac{a-c}{2b}$.
Why does a cartel break down?	Each member gains by producing more: $\Delta\pi_A = -\,q_B\,\Delta P > 0$.
What is a dominant strategy?	The best choice whatever the rival does.
What is a Nash equilibrium?	Each choice is best given the other's choice.
Where does oligopoly output lie?	Between monopoly and perfect competition.
Is collusion legal?	No: price-fixing is illegal under the Sherman Act.

Appendix A.1: the algebra behind the two areas

Joint profit is $\pi = (P - c)\,Q$. Output rises by ΔQ; the price changes by $\Delta P < 0$; P is the new price.

$\Delta\pi = (P - c)\,\Delta Q + Q\,\Delta P$	gain plus loss
$0 = (P - c)\,\Delta Q + Q\,\Delta P$	cartel output: gain equals loss
$0 = (P - c)\,\Delta Q + q_A\,\Delta P + q_B\,\Delta P$	replace Q with $q_A + q_B$
$\Delta\pi_A = (P - c)\,\Delta Q + q_A\,\Delta P$	A alone: all the gain, its own part of the loss
$\Delta\pi_A = -\,q_B\,\Delta P > 0$	subtract the third line from the fourth

Dividing the second line by ΔQ gives $P + Q\,\dfrac{\Delta P}{\Delta Q} = c$, that is, MR = MC.

Appendix A.2: the cement arithmetic

Demand $P = 200 - Q/10$; $c = \$20$; agreed output 450 tons each.

- Cartel: $Q = 900$, $P = 200 - 90 = \$110$; each earns $(110 - 20) \times 450 = \$40{,}500$.
- A cheats: $Q = 600 + 450 = 1{,}050$, $P = 200 - 105 = \$95$.
- A earns $(95 - 20) \times 600 = \$45{,}000$; B earns $(95 - 20) \times 450 = \$33{,}750$.
- A's gain: output effect $(95 - 20) \times 150 = \$11{,}250$; price effect on A's 450 tons $450 \times (-15) = -\$6{,}750$; net $+\$4{,}500$.
- B's loss: $450 \times (-15) = -\$6{,}750$. Joint profit falls by \$2,250: a 150-ton jump overshoots the one-unit rule, but the cheat still gains.
- Both cheat: $Q = 1{,}200$, $P = \$80$; each earns $(80 - 20) \times 600 = \$36{,}000$.

Appendix A.3: where the firms end up

Firm A takes B's output q_B as given.

$P = (a - b\,q_B) - b\,q_A$	the demand left for A: a straight line, slope $-b$
$\text{MR}_A = (a - b\,q_B) - 2b\,q_A$	MR is twice as steep (Chapter 16)
$c = a - b\,q_B - 2b\,q_A$	set $\text{MR}_A = \text{MC}$
$c = a - 3bq$	equal outputs, $q_A = q_B = q$
$q = \dfrac{a-c}{3b}$	solve for q

Cartel share: half of $\frac{a-c}{2b}$, that is $\frac{a-c}{4b}$. Cement: $q = \frac{180}{3/10} = 600$ tons; cartel share 450 tons.